INCH-POUND
MIL-C-19978/19C
18 June 2001
SUPERSEDING
MIL-C-19978/19B
18 April 1995

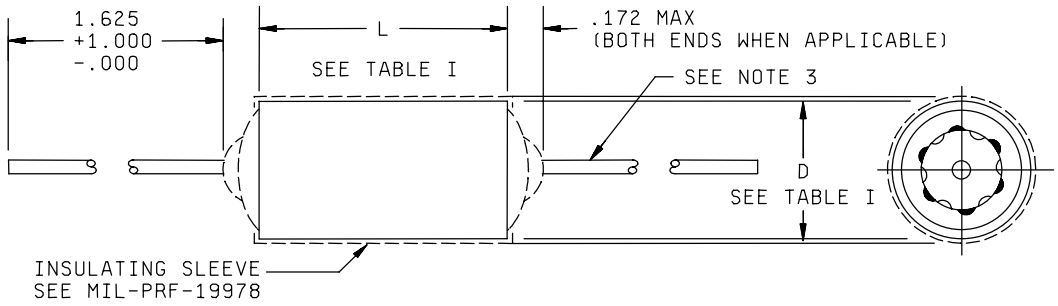
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, TUBULAR (INSULATED), (HERMETICALLY SEALED IN METAL CASES), STYLE CQ05

INACTIVE FOR NEW DESIGN
AFTER 30 AUGUST 1974

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-19978.



Inches	mm	Inches	mm
.001	0.03	.312	7.92
.004	0.10	.400	10.16
.025	0.64	.562	14.27
.032	0.81	1.000	25.40
.172	4.37	1.438	36.53
.235	5.97	1.625	41.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Leads shall be of tinned solid wire, .025 (No. 22 AWG) for cases .235 and .312 in diameter; and .032 (No. 20 AWG) for cases .400 diameter and above. Tolerance on all lead wire diameters shall be + .004, -.001.
4. Capacitors with dimension L. of 1.438 or D of .562 and larger, are not intended to be supported by their leads. These capacitors shall be supported with a supplementary means of mounting, such as a wrap-around band. The supporting device will not be supplied with the capacitor.
5. Lead length may be a minimum of 1 inch long for use in tape and reel packaging when specified in the ordering data.

FIGURE 1. Style CQ05 capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

TABLE I. Style CQ05 capacitors.

Part or Identifying Number (PIN) 1/	DC voltage rating (volts)	Capacitance (µF)	Case dimensions (inches) 2/		
				Circuit 1	Circuit 3
			D+.010 -.020	L ±.031	L ±.031
CQ05A-KB273K3-	100	.027	0.235	0.812	0.750
CQ05A-KB333K3-	100	.033	0.235	0.812	0.750
CQ05A-KB823K3-	100	.082	0.312	0.938	0.875
CQ05A-KB104K3-	100	.10	0.312	0.938	0.875
CQ05A-KB124K3-	100	.12	0.400	0.938	0.875
CQ05A-KB154K3-	100	.15	0.400	0.938	0.875
CQ05A-KB184K3-	100	.18	0.400	1.188	1.125
CQ05A-KB224K3-	100	.22	0.400	1.188	1.125
CQ05A-KB274K3-	100	.27	0.400	1.438	1.375
CQ05A-KB334K3-	100	.33	0.400	1.438	1.375
CQ05A-KB394K3-	100	.39	0.562	1.188	1.125
CQ05A-KB474K3-	100	.47	0.562	1.188	1.125
CQ05A-KB564K3-	100	.56	0.562	1.438	1.375
CQ05A-KB684K3-	100	.68	0.562	1.438	1.375
CQ05A-KB824K3-	100	.82	0.562	1.938	1.875
CQ05A-KB105K3-	100	1.00	0.562	1.938	1.875
CQ05A-KC123K3-	200	.012	0.235	0.812	0.750
CQ05A-KC153K3-	200	.015	0.235	0.812	0.750
CQ05A-KC183K3-	200	.018	0.235	0.812	0.750
CQ05A-KC223K3-	200	.022	0.235	0.812	0.750
CQ05A-KC393K3-	200	.039	0.312	0.938	0.875
CQ05A-KC473K3-	200	.047	0.312	0.938	0.875
CQ05A-KC563K3-	200	.056	0.312	0.938	0.875
CQ05A-KC683K3-	200	.068	0.312	0.938	0.875
CQ05A-KC823K3-	200	.082	0.400	0.938	0.875
CQ05A-KC104K3-	200	.10	0.400	0.938	0.875
CQ05A-KC124K3-	200	.12	0.400	1.188	1.125
CQ05A-KC154K3-	200	.15	0.400	1.188	1.125
CQ05A-KC184K3-	200	.18	0.400	1.438	1.375
CQ05A-KC224K3-	200	.22	0.400	1.438	1.375
CQ05A-KC274K3-	200	.27	0.562	1.188	1.125
CQ05A-KC334K3-	200	.33	0.562	1.188	1.125
CQ05A-KC394K3-	200	.39	0.562	1.438	1.375
CQ05A-KC474K3-	200	.47	0.562	1.438	1.375
CQ05A-KC564K3-	200	.56	0.562	1.938	1.875
CQ05A-KC684K3-	200	.68	0.562	1.938	1.875
CQ05A-KC824K3-	200	.82	0.670	1.938	1.875
CQ05A-KC105K3-	200	1.00	0.670	1.938	1.875
CQ05A-KE822K3-	400	.0082	0.235	0.812	0.750
CQ05A-KE103K3-	400	.010	0.235	0.812	0.750
CQ05A-KE273K3-	400	.027	0.312	0.938	0.875
CQ05A-KE333K3-	400	.033	0.312	0.938	0.875
CQ05A-KE393K3-	400	.039	0.400	0.938	0.875
CQ05A-KE473K3-	400	.047	0.400	0.938	0.875
CQ05A-KE563K3-	400	.056	0.400	1.188	1.125
CQ05A-KE683K3-	400	.068	0.400	1.188	1.125

See footnotes at end of table.

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TABLE I. Style CQ05 capacitors. - Continued.

Part or Identifying Number (PIN) 1/	DC voltage rating (volts)	Capacitance (µF)	Case dimensions (inches) 2/		
			D+.010 -.020	Circuit 1 L ±.031	Circuit 3 L ±.031
CQ05A-KE823K3-	400	.082	0.400	1.438	1.375
CQ05A-KE104K3-	400	.100	0.400	1.438	1.375
CQ05A-KE124K3-	400	.120	0.562	1.188	1.125
CQ05A-KE154K3-	400	.150	0.562	1.188	1.125
CQ05A-KE184K3-	400	.180	0.562	1.438	1.375
CQ05A-KE224K3-	400	.220	0.562	1.438	1.375
CQ05A-KE274K3-	400	.270	0.562	1.938	1.875
CQ05A-KE334K3-	400	.330	0.562	1.938	1.875
CQ05A-KE394K3-	400	.390	0.670	1.938	1.875
CQ05A-KE474K3-	400	.470	0.670	1.938	1.875
CQ05A-KE564K3-	400	.560	0.750	2.188	2.125
CQ05A-KE684K3-	400	.680	0.750	2.188	2.125
CQ05A-KE824K3-	400	.820	1.000	1.938	1.875
CQ05A-KE105K3-	400	1.000	1.000	1.938	1.875
CQ05A-KF102K3-	600	.001	0.235	0.812	0.750
CQ05A-KF122K3-	600	.0012	0.235	0.812	0.750
CQ05A-KF152K3-	600	.0015	0.235	0.812	0.750
CQ05A-KF182K3-	600	.0018	0.235	0.812	0.750
CQ05A-KF222K3-	600	.0022	0.235	0.812	0.750
CQ05A-KF272K3-	600	.0027	0.235	0.812	0.750
CQ05A-KF332K3-	600	.0033	0.235	0.812	0.750
CQ05A-KF392K3-	600	.0039	0.235	0.812	0.750
CQ05A-KF472K3-	600	.0047	0.235	0.812	0.750
CQ05A-KF562K3-	600	.0056	0.235	0.812	0.750
CQ05A-KF682K3-	600	.0068	0.235	0.812	0.750
CQ05A-KF822K3-	600	.0082	0.312	0.938	0.875
CQ05A-KF103K3-	600	.010	0.312	0.938	0.875
CQ05A-KF123K3-	600	.012	0.312	0.938	0.875
CQ05A-KF153K3-	600	.015	0.312	0.938	0.875
CQ05A-KF183K3-	600	.018	0.312	0.938	0.875
CQ05A-KF223K3-	600	.022	0.312	0.938	0.875
CQ05A-KF273K3-	600	.027	0.400	0.938	0.875
CQ05A-KF333K3-	600	.033	0.400	0.938	0.875
CQ05A-KF393K3-	600	.039	0.400	1.188	1.125
CQ05A-KF473K3-	600	.047	0.400	1.188	1.125
CQ05A-KF563K3-	600	.056	0.400	1.438	1.375
CQ05A-KF683K3-	600	.068	0.400	1.438	1.375
CQ05A-KF823K3-	600	.082	0.562	1.188	1.125
CQ05A-KF104K3-	600	.10	0.562	1.188	1.125
CQ05A-KF124K3-	600	.12	0.562	1.438	1.375
CQ05A-KF154K3-	600	.15	0.562	1.438	1.375
CQ05A-KF184K3-	600	.18	0.562	1.938	1.875
CQ05A-KF224K3-	600	.22	0.562	1.938	1.875
CQ05A-KF274K3-	600	.27	0.562	1.938	1.875
CQ05A-KF334K3-	600	.33	0.562	1.938	1.875
CQ05A-KF394K3-	600	.39	0.750	2.188	2.125
CQ05A-KF474K3-	600	.47	0.750	2.188	2.125
CQ05A-KF564K3-	600	.56	1.000	1.938	1.875
CQ05A-KF684K3-	600	.68	1.000	1.938	1.875
CQ05A-KG153K3-	1000	.015	0.400	0.938	0.875

See footnotes at end of table.

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TABLE I. Style CQ05 capacitors. - Continued.

Part or Identifying Number (PIN) 1/	DC voltage rating (volts)	Capacitance (µF)	Case dimensions (inches) 2/		
			D+.010 -.020	Circuit 1 L ±.031	Circuit 3 L ±.031
CQ05A-KG223K3-	1000	0.022	0.400	1.188	1.125
CQ05A-KG273K3-	1000	0.027	0.400	1.438	1.375
CQ05A-KG333K3-	1000	0.033	0.400	1.438	1.375
CQ05A-KG473K3-	1000	0.047	0.562	1.188	1.125
CQ05A-KG563K3-	1000	0.056	0.562	1.438	1.375
CQ05A-KG683K3-	1000	0.068	0.562	1.438	1.375
CQ05A-KG823K3-	1000	0.082	0.562	1.688	1.625
CQ05A-KG104K3-	1000	0.10	0.562	1.688	1.625
CQ05A-KG124K3-	1000	0.12	0.670	1.688	1.625
CQ05A-KG154K3-	1000	0.15	0.670	1.688	1.625
CQ05A-KG224K3-	1000	0.22	0.750	2.188	2.125
CQ05A-KG274K3-	1000	0.27	0.750	2.688	2.625
CQ05A-KG334K3-	1000	0.33	0.750	2.688	2.625
CQ05A-KG474K3-	1000	0.47	1.000	2.188	2.125

1/ Complete PIN shall include additional symbol to indicate circuit.
2/ See table II for metric equivalents.

TABLE II. Millimeter equivalent of decimal inches.

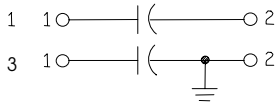
Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
0.010	0.25	0.400	10.16	0.875	22.23	1.375	34.93	1.938	49.23
0.020	0.51	0.562	14.27	0.938	23.83	1.438	36.53	2.125	53.98
0.031	0.79	0.670	17.02	1.000	25.40	1.625	41.28	2.188	55.58
0.235	5.97	0.750	19.05	1.125	28.58	1.688	42.88	2.625	66.68
0.312	7.92	0.812	20.62	1.188	30.18	1.875	47.63	2.688	68.28

Part or Identifying Number (PIN): See table I.

Case material: Nonmagnetic (end seal may be of magnetic material).

Dielectric material: Paper and polyethelene terephthalate (K).

Circuit diagram:



Terminals: Axial-wire lead (see figure 1).

Rated voltage: See table I.

Rated temperature: -65° to +125°C.

Capacitance (nominal): See table I.

Capacitance tolerance: ±10 percent (symbol K).

Dissipation factor (DF) (max): 1.0 percent.

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Seal: Method 112 of MIL-STD-202, test condition letter A.

Dielectric withstanding voltage (DWV):

Sleeving: In accordance with MIL-PRF-19978.

Barometric pressure, qualification only: In accordance with MIL-PRF-19978.

Test points:

Circuit diagram 1: Between terminals and case.

Circuit diagram 3: Between ungrounded terminal and case.

Insulation resistance (IR):

Sleeving: In accordance with MIL-PRF-19978.

Terminal to terminal: See table III.

TABLE III. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
	At 25°C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Terminal to case: Greater than 10,000 megohms.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:

Direction and duration of motion: 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-PRF-19978.

Immersion:

DWV:

Sleeving: In accordance with MIL-PRF-19978.

IR:

Sleeving: In accordance with MIL-PRF-19978.

Solderability: In accordance with MIL-PRF-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter D.

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Moisture resistance:

DWV:

Sleeving: In accordance with MIL-PRF-19978.

IR:

Sleeving: In accordance with MIL-PRF-19978.

Stability at low and high temperature:

Low temperature:

Test temperature: -65°C +0°, -5°C.

Capacitance change (max): -10 percent.

High temperature:

Test temperature: +125°C +5°, -0°C.

Capacitance change (max): +10 percent.

Life:

Capacitance change (max): ±5 percent.

Resistance to soldering heat: In accordance with MIL-PRF-19978.

Substitution data: See table IV.

NOTE: Style CQ05 capacitors are substitutable for styles CP04 and CP05 capacitors formerly covered in MIL-C-25/1K.

TABLE IV. Substitutability data.

From	To	From	To
MIL-C-25/1K (CP04 and CP05)	MIL-C-19978/19A (CQ05)	MIL-C-25/1K (CP04 and CP05)	MIL-C-19978/19A (CQ05)
CP0-A-KB273K-	CQ05A-KB273K3-	CP0-A-KC223K-	CQ05A-KC223K3-
CP0-A-KB333K-	CQ05A-KB333K3-	CP0-A-KC393K-	CQ05A-KC393K3-
CP0-A-KB823K-	CQ05A-KB823K3-	CP0-A-KC473K-	CQ05A-KC473K3-
CP0-A-KB104K-	CQ05A-KB104K3-	CP0-A-KC563K-	CQ05A-KC563K3-
CP0-A-KB124K-	CQ05A-KB124K3-	CP0-A-KC683K-	CQ05A-KC683K3-
CP0-A-KB154K-	CQ05A-KB154K3-	CP0-A-KC823K-	CQ05A-KC823K3-
CP0-A-KB184K-	CQ05A-KB184K3-	CP0-A-KC104K-	CQ05A-KC104K3-
CP0-A-KB224K-	CQ05A-KB224K3-	CP0-A-KC124K-	CQ05A-KC124K3-
CP0-A-KB274K-	CQ05A-KB274K3-	CP0-A-KC154K-	CQ05A-KC154K3-
CP0-A-KB334K-	CQ05A-KB334K3-	CP0-A-KC184K-	CQ05A-KC184K3-
CP0-A-KB394K-	CQ05A-KB394K3-	CP0-A-KC224K-	CQ05A-KC224K3-
CP0-A-KB474K-	CQ05A-KB474K3-	CP0-A-KC274K-	CQ05A-KC274K3-
CP0-A-KB564K-	CQ05A-KB564K3-	CP0-A-KC334K-	CQ05A-KC334K3-
CP0-A-KB684K-	CQ05A-KB684K3-	CP0-A-KC394K-	CQ05A-KC394K3-

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TABLE IV. Substitutability data - Continued.

From	To	From	To
MIL-C-25/1K (CP04 and CP05)	MIL-C-19978/19A (CQ05)	MIL-C-25/1K (CP04 and CP05)	MIL-C-19978/19A (CQ05)
CP0-A-KB824K-	CQ05A-KB824K3-	CP0-A-KC474K-	CQ05A-KC474K3-
CP0-A-KB105K-	CQ05A-KB105K3-	CP0-A-KC564K-	CQ05A-KC564K3-
CP0-A-KC123K-	CQ05A-KC123K3-	CP0-A-KC684K-	CQ05A-KC684K3-
CP0-A-KC153K-	CQ05A-KC153K3-	CP0-A-KC824K-	CQ05A-KC824K3-
CP0-A-KC183K-	CQ05A-KC183K3-	CP0-A-KC105K-	CQ05A-KC105K3-
CP0-A-KE822K-	CQ05A-KE822K3-	CP0-A-KF153K-	CQ05A-KF153K3-
CP0-A-KE103K-	CQ05A-KE103K3-	CP0-A-KF183K-	CQ05A-KF183K3-
CP0-A-KE273K-	CQ05A-KE273K3-	CP0-A-KF223K-	CQ05A-KF223K3-
CP0-A-KE333K-	CQ05A-KE333K3-	CP0-A-KF273K-	CQ05A-KF273K3-
CP0-A-KE393K-	CQ05A-KE393K3-	CP0-A-KF333K-	CQ05A-KF333K3-
CP0-A-KE473K-	CQ05A-KE473K3-	CP0-A-KF393K-	CQ05A-KF393K3-
CP0-A-KE563K-	CQ05A-KE563K3-	CP0-A-KF473K-	CQ05A-KF473K3-
CP0-A-KE683K-	CQ05A-KE683K3-	CP0-A-KF563K-	CQ05A-KF563K3-
CP0-A-KE823K-	CQ05A-KE823K3-	CP0-A-KF683K-	CQ05A-KF683K3-
CP0-A-KE104K-	CQ05A-KE104K3-	CP0-A-KF823K-	CQ05A-KF823K3-
CP0-A-KE124K-	CQ05A-KE124K3-	CP0-A-KF104K-	CQ05A-KF104K3-
CP0-A-KE154K-	CQ05A-KE154K3-	CP0-A-KF124K-	CQ05A-KF124K3-
CP0-A-KE184K-	CQ05A-KE184K3-	CP0-A-KF154K-	CQ05A-KF154K3-
CP0-A-KE224K-	CQ05A-KE224K3-	CP0-A-KF184K-	CQ05A-KF184K3-
CP0-A-KE274K-	CQ05A-KE274K3-	CP0-A-KF224K-	CQ05A-KF224K3-
CP0-A-KE334K-	CQ05A-KE334K3-	CP0-A-KF274K-	CQ05A-KF274K3-
CP0-A-KE394K-	CQ05A-KE394K3-	CP0-A-KF334K-	CQ05A-KF334K3-
CP0-A-KE474K-	CQ05A-KE474K3-	CP0-A-KF394K-	CQ05A-KF394K3-
CP0-A-KE564K-	CQ05A-KE564K3-	CP0-A-KF474K-	CQ05A-KF474K3-
CP0-A-KE684K-	CQ05A-KE684K3-	CP0-A-KF564K-	CQ05A-KF564K3-
CP0-A-KE824K-	CQ05A-KE824K3-	CP0-A-KF684K-	CQ05A-KF684K3-
CP0-A-KE105K-	CQ05A-KE105K3-	CP0-A-KG153K-	CQ05A-KG153K3-
CP0-A-KF102K-	CQ05A-KF102K3-	CP0-A-KG223K-	CQ05A-KG223K3-
CP0-A-KF122K-	CQ05A-KF122K3-	CP0-A-KG273K-	CQ05A-KG273K3-
CP0-A-KF152K-	CQ05A-KF152K3-	CP0-A-KG333K-	CQ05A-KG333K3-
CP0-A-KF182K-	CQ05A-KF182K3-	CP0-A-KG473K-	CQ05A-KG473K3-
CP0-A-KF222K-	CQ05A-KF222K3-	CP0-A-KG563K-	CQ05A-KG563K3-
CP0-A-KF272K-	CQ05A-KF272K3-	CP0-A-KG683K-	CQ05A-KG683K3-
CP0-A-KF332K-	CQ05A-KF332K3-	CP0-A-KG823K-	CQ05A-KG823K3-
CP0-A-KF392K-	CQ05A-KF392K3-	CP0-A-KG104K-	CQ05A-KG104K3-
CP0-A-KF472K-	CQ05A-KF472K3-	CP0-A-KG124K-	CQ05A-KG124K3-
CP0-A-KF562K-	CQ05A-KF562K3-	CP0-A-KG154K-	CQ05A-KG154K3-
CP0-A-KF682K-	CQ05A-KF682K3-	CP0-A-KG224K-	CQ05A-KG224K3-
CP0-A-KF822K-	CQ05A-KF822K3-	CP0-A-KG274K-	CQ05A-KG274K3-
CP0-A-KF103K-	CQ05A-KF103K3-	CP0-A-KG334K-	CQ05A-KG334K3-
CP0-A-KF123K-	CQ05A-KF123K3-	CP0-A-KG474K-	CQ05A-KG474K3-

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, AT, AV, MI
Navy - AS, MC, OS, SH, TD
Air Force - 19

Preparing activity:
DLA - CC

(Project 5910-2118-05)

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